

Checkpoint 1: When Is a Positive Really Positive?

! Important

Due 10/18 (12 points). See [Project Central](#) for the full project overview. Make sure you have addressed any feedback from Checkpoint 0.

Goal

Use conditional probability and Bayes' rule to answer: *If someone in my scenario gets a positive result, how likely is it that they truly have the condition?*

Course topics used: events, conditional probability, the Law of Total Probability, Bayes' rule, independence.

Tasks

1. Set up the problem

- Define your events with notation (e.g., D = has the disease, T^+ = tests positive).
- Write each input value from Checkpoint 0 as a probability statement, for example $P(T^+ | D) = 0.90$. Cite your sources.
- Draw a tree diagram **or** a 2×2 table for a hypothetical population of 100,000 people.

2. Solve it analytically

- Use the Law of Total Probability to find $P(T^+)$.
- Use Bayes' rule to find the **positive predictive value** $P(D | T^+)$ and the **negative predictive value** $P(D^c | T^-)$.
- Show your work. You may use R as a calculator, but write out each formula first.

3. Simulate it

- Simulate a population of at least 100,000 people in R. Assign each person a disease status, then a test result that depends on their disease status.
- Estimate PPV and NPV from your simulated population and compare them with your analytic answers.
- In 1–2 sentences, explain why the simulated and analytic values are close but not identical.

Starter code (replace the input values with your own):

```
set.seed(550)
N <- 1e5
prev <- 0.01 # P(D)
sens <- 0.90 # P(T+ | D)
spec <- 0.95 # P(T- | D^c)

disease <- rbinom(N, size = 1, prob = prev)
test_pos <- ifelse(disease == 1,
                   rbinom(N, size = 1, prob = sens),
                   rbinom(N, size = 1, prob = 1 - spec))

ppv_sim <- mean(disease[test_pos == 1])
ppv_bayes <- (sens * prev) / (sens * prev + (1 - spec) * (1 - prev))

c(simulated = ppv_sim, analytic = ppv_bayes)
```

4. Explore: how does prevalence matter?

- Compute the analytic PPV for prevalence values from 0.1% to 50%, keeping sensitivity and specificity fixed.
- Plot PPV (y-axis) against prevalence (x-axis), and mark your scenario's prevalence on the plot.
- In 2–3 sentences, describe what happens to PPV as prevalence changes. When would this test be most useful?

5. Say it

Write a short paragraph (4–6 sentences) **for a patient** who just received a positive result. Explain what the result means using natural frequencies (e.g., “Out of 1,000 people like you who test positive, about ____ actually have...”) rather than probability notation.

Submission

Submit a rendered HTML file (use `code-fold: true`) and your `.qmd`. One submission per team.

Rubric (12 points)

Criterion	3	2	1
Setup (notation, input values, tree or table)	Clear and correct; inputs cited	Minor gaps in notation or citation	Major errors or missing
Analytic solution (LTP, PPV, NPV)	Correct, with work shown	Small errors	Mostly incorrect or no work shown
Simulation (code, comparison to analytic answer, prevalence plot)	Agrees with analytic answer; differences explained; plot is clear	Runs, but comparison or plot is weak	Missing or incorrect
Interpretation (prevalence discussion, patient paragraph)	Accurate and in plain language	Somewhat technical or vague	Missing or misleading